

Presentation 2

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://docs.google.com/presentation/d/1fGIhBWiy8Le5p4dEeeG4pNLmgn7Y6FLq0of-rLRMNc/edit?slide=id.g39ba5113368_1_2#slide=id.g39ba5113368_1_2

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Bonsitu_Kebeto_LegalIssuesAnalysis_10_21_Code.ipynb

Dashboard:

https://app.powerbi.com/links/Cu5kaNhr6w?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

https://app.powerbi.com/links/vL1hBdfPoB?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

https://app.powerbi.com/links/f0_UTfsrwH?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

• What did you do?

1. Analyzed overall call frequency across 10 legal menu categories and applied a 5% threshold to identify low-volume categories that could be consolidated. I grouped the data by legal category and counted unique calls using "Contact Session ID", then calculated what percentage of total calls each category represented.
2. Created a monthly comparison graph showing call volume trends across 2024-2025). I grouped the data by month and aggregated total calls across all legal categories to identify seasonal patterns.
3. Examined day-of-week patterns by analyzing Monday through Friday call volumes for the top legal categories (those above the 5% threshold). I used a standardized metric—Calls Per Day, which is the average of typical daily call rates.

• How does it help the project?

The analysis identifies low-volume categories for menu consolidation using a 5% threshold, reveals seasonal demand peaks in January and April through monthly trends, and shows that Mondays consistently have 20-25% more calls than Fridays across all legal categories.

• Issues faced (if any)

N/A

• Attempts to resolve issues (if any)

N/A

• Issues resolved (if any)

N/A

• Next steps

Create a drill-down of call patterns within sub-menus for Family, Housing, and Other Legal menu.
Examine call distribution by time of day to identify peak calling hours.

- References (Mention if you built up on someone else's work)

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Presentation 1

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_9Oct_Jasmine.pdf

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Bonsitu_Kebeto_Presentation1_Code.ipynb

Dashboard:

https://app.powerbi.com/links/I-JSupMipK?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare&bookmarkGuid=06c57976-3112-472b-b5b4-964a396b5eba
https://app.powerbi.com/links/goOOGrYVjf?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

- What did you do?

I created two graphs, the first showing how call volume varied over time across intake lines. I did this by grouping the CAR dataset by month and intake line, and plotting the unique calls using "Contact Session ID".

The second graph looks at the peak call hours for the "Main Number Telephony EP" intake line, by first filtering the dataset for that specific intake line and grouping by hour, then counting the unique calls using "Contact Session ID".

I then exported the results to a csv and used that to create visualizations for both in Power BI.

- How does it help the project?

The first graph was to identify the intake lines with the most and least amount of traffic so that Legal Aid could remove the options that get the least amount of traffic to make intake simpler for the caller and combine it with other intake lines.

The second graph was in order to see which hours were busiest for their most used intake line so that Legal Aid could prioritize staffing during those hours.

- Issues faced (if any)

There were some questions I had to clarify about the difference between intake lines and the 6 numbers that customers access legal aid from.

- Attempts to resolve issues (if any)

I was able to clarify the questions I have with the documentation on Github and revise my recommendations.

- Issues resolved (if any)

I updated the presentation slides to be more specific about consolidating all of the various Legal Issues intakes lines into just one.

- Next steps

I want to use the All Calls dataset to calculate the duration of the calls and abandonment rate for each intake line to see the performance across different intake lines in order to further justify which specific lines should be combined or discontinued.

- References (Mention if you built up on someone else's work)

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